

A missing sharpness edge for spherical maximal functions on H-type groups with center dimension three or seven

Automated mathematical discovery run `fa_banach_001`

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Status. Candidate substantial partial result, likely valid, subject to human review. We prove the missing Q_3Q_4 necessary inequality for untilted H-type groups whose center has dimension $m = 3$ or $m = 7$, and more generally under an explicit common-invariant-block hypothesis. General Métivier groups and arbitrary tilts are not resolved.

1 Source problem

The source is J. Roos, A. Seeger, and R. Srivastava, *Lebesgue space estimates for spherical maximal functions on Heisenberg groups*, arXiv:2103.09734v2 [1]. In the notation of that paper, the horizontal dimension is $h = 2n$, the center dimension is m , and the total Euclidean dimension is

$$D = h + m.$$

For skew-symmetric $h \times h$ matrices $J_1, \dots, J_m$, write $x^\top \vec{J}\omega = (x^\top J_1\omega, \dots, x^\top J_m\omega)$. If $\Lambda : \mathbb{R}^h \rightarrow \mathbb{R}^m$ is the tilt, the local maximal operator considered in the source is

$$\mathcal{M}f(x, z) = \sup_{1 \leq t \leq 2} \left| \int_{S^{h-1}} f(x - t\omega, z - tx^\top \vec{J}\omega - t^2 \Lambda\omega) d\sigma(\omega) \right|. \quad (1)$$

The source proves positive estimates in a quadrilateral with vertices

$$\begin{aligned} Q_3 &= \left(\frac{D-1}{D+m}, \frac{m+1}{D+m} \right), \\ Q_4 &= \left(\frac{D(D-1)}{\Delta}, \frac{(m+1)(D-1)}{\Delta} \right), \quad \Delta = D^2 + (D+1)m + 1. \end{aligned} \quad (2)$$

Its Remark 1.2(iv), on source p. 5, asks for analogues when $m \geq 2$ of its new example proving sharpness of the line Q_3Q_4 for $m = 1$:

on variable coefficient Nikodym estimates in [22].

(iii) For $m+3 \leq 2n$, one can use an alternative approach to the $L^{q'} \rightarrow L^q$ estimates which does not require assumption (1.4), cf. Remark 5.1. One also obtains the endpoint restricted weak type estimate for the point Q_2 provided that $m+3 < 2n$.

(iv) The example in §6.5 demonstrating the sharpness of the line Q_3Q_4 in the case $m = 1$ seems to be new. It would be interesting to see whether there exists similar examples for $m \geq 2$ and to settle the problem of sharpness for those cases. It would also be interesting to analyze what happens when the size restriction (1.4) on Λ is dropped.

L^p improving estimates for spherical averages. We now discuss another problem considered in [3], concerning sparse bounds for the lacunary spherical

In particular, it asks whether similar examples exist for $m \geq 2$ and whether the corresponding sharpness problem can be settled.

2 Proof intuition

The source's $m = 1$ construction singles out a two-plane invariant under the single complex structure J . Horizontal directions in that plane are localized at scale δ , directions in its orthogonal complement at scale $\sqrt{\delta}$, and the center at scale δ . The apparent $\sqrt{\delta}$ error in the group law becomes a product of two $\sqrt{\delta}$ errors, hence has size δ .

The same mechanism works for center dimension m if all J_i preserve a common $(m + 1)$ -dimensional horizontal block. The resulting volume exponents give exactly the line through the published Q_3 and Q_4 . H-type groups are real Clifford modules. In the exceptional center dimensions 3 and 7, their irreducible Clifford blocks have dimensions 4 and 8—exactly $m + 1$ —so the construction applies. The block hypothesis is also the limitation: a general skew pencil need not have such a subspace, and a general tilt can see the coarse $\sqrt{\delta}$ directions.

3 Invariant-block Knapp theorem

Theorem 1. *Suppose there is a common invariant subspace $V \subset \mathbb{R}^h$ for $J_1, \dots, J_m$ with dimension r . Assume also*

$$\Lambda|_{V^\perp} = 0. \quad (3)$$

If (1) is bounded from $L^p(\mathbb{R}^{h+m})$ to $L^q(\mathbb{R}^{h+m})$, then

$$D - m + r - 2 + \frac{D + m - r}{q} \geq \frac{D + m + r}{p}. \quad (4)$$

In particular, if $r = m + 1$, boundedness forces

$$(D - 1) \left(1 + \frac{1}{q} \right) \geq \frac{D + 2m + 1}{p}. \quad (5)$$

Equality in (5) is precisely the line through the two points Q_3, Q_4 in (2).

Proof. Let P and $P^\perp$ be the orthogonal projections onto V and $V^\perp$. Since every J_i is skew-symmetric and preserves V , it also preserves $V^\perp$. Fix a small $\delta > 0$ and, with a sufficiently large constant C , set

$$E_\delta = \{(y, w) : |Py| \leq C\delta, \quad |P^\perp y| \leq C\delta^{1/2}, \quad |w| \leq C\delta\}, \quad f_\delta = \mathbf{1}_{E_\delta}.$$

Then

$$\|f_\delta\|_p \asymp \delta^{\{(h-r)/2+r+m\}/p} = \delta^{(D+m+r)/(2p)}. \quad (6)$$

Choose a relatively compact coordinate patch U of the unit sphere in V . Consider the output set

$$R_\delta = \{(x, z) : 9/8 \leq |Px| \leq 15/8, \quad Px/|Px| \in U, \quad |P^\perp x| \leq c\delta^{1/2}, \quad |z - |Px|\Lambda x| \leq c\delta\}, \quad (7)$$

where $c > 0$ is a sufficiently small fixed constant. Fubini's theorem in orthogonal coordinates gives

$$|R_\delta| \asymp \delta^{(h-r)/2+m} = \delta^{(D+m-r)/2}. \quad (8)$$

Fix $(x, z) \in R_\delta$, put $t = |Px|$, and let $v = Px/t$. In local sphere coordinates define $\Omega_x \subset S^{h-1}$ by requiring

$$|P^\perp \omega| \leq c\delta^{1/2} \quad \text{and} \quad \text{dist}_{S(V)} \left(\frac{P\omega}{|P\omega|}, v \right) \leq c\delta.$$

There are $h - r$ independent coordinates of size $\delta^{1/2}$ and $r - 1$ independent coordinates of size δ . Hence

$$\sigma(\Omega_x) \asymp \delta^{(h-r)/2+r-1} = \delta^{(D-m+r-2)/2}. \quad (9)$$

Moreover, for $\omega \in \Omega_x$, the unit-sphere equation gives

$$|P(x - t\omega)| \lesssim \delta, \quad |P^\perp(x - t\omega)| \lesssim \delta^{1/2}. \quad (10)$$

For each i , skew-symmetry yields

$$x^\top J_i \omega = (x - t\omega)^\top J_i \omega.$$

Because V and $V^\perp$ are invariant, the mixed terms in the orthogonal decomposition vanish. Thus (10) gives

$$\begin{aligned} |x^\top J_i \omega| &\leq |P(x - t\omega)| \|J_i\| |P\omega| \\ &\quad + |P^\perp(x - t\omega)| \|J_i\| |P^\perp \omega| \lesssim \delta. \end{aligned}$$

Using $t = |Px|$ and (3), the vertical argument in (1) can be rearranged as

$$z - tx^\top \vec{J}\omega - t^2 \Lambda \omega = z - t\Lambda x + t\Lambda(x - t\omega) - tx^\top \vec{J}\omega,$$

and every term on the right is $O(\delta)$. Enlarging C if necessary, the full argument of f_δ lies in E_δ for all $\omega \in \Omega_x$. Consequently

$$\mathcal{M}f_\delta(x, z) \gtrsim \sigma(\Omega_x) \asymp \delta^{(D-m+r-2)/2}, \quad (x, z) \in R_\delta. \quad (11)$$

Combining (6), (8), and (11), an $L^p \rightarrow L^q$ estimate independent of δ requires

$$\frac{D - m + r - 2}{2} + \frac{D + m - r}{2q} \geq \frac{D + m + r}{2p},$$

which is (4).

If $r = m + 1$, this becomes (5). Direct substitution of (2) gives equality at both Q_3 and Q_4 ; equivalently, the equality line is

$$(D - 1)(1 + 1/q) = (D + 2m + 1)/p.$$

□

4 H-type consequence

Recall that for an H-type group one may choose an orthonormal basis of the center so that

$$J_i J_j + J_j J_i = -2\delta_{ij} I. \quad (12)$$

Thus the horizontal space is a finite-dimensional real module over the Clifford algebra $\text{Cl}_{0,m}$; see Kaplan [2].

Corollary 2. *Let G be an H-type group with center dimension $m = 3$ or $m = 7$. For the untilted spherical maximal operator ($\Lambda = 0$), $L^p \rightarrow L^q$ boundedness implies the exact $Q_3 Q_4$ necessary inequality (5).*

The same conclusion holds for a tilt Λ whenever there is an irreducible Clifford block V such that $\text{range } \Lambda^\top \subset V$.

Proof. The real Clifford-algebra classification gives

$$\mathrm{Cl}_{0,3} \cong \mathbb{H} \oplus \mathbb{H}, \quad \mathrm{Cl}_{0,7} \cong M_8(\mathbb{R}) \oplus M_8(\mathbb{R}).$$

Accordingly, every finite-dimensional real module is an orthogonal direct sum of invariant irreducible modules of real dimension 4 when $m = 3$ and 8 when $m = 7$ [3]. Choose one such block V . In both cases $\dim V = m + 1$. For $\Lambda = 0$, condition (3) is automatic. More generally, $\mathrm{range} \Lambda^\top \subset V$ is equivalent to that condition. Theorem 1 applies. $\square$

Together with the other necessary conditions already proved in the source, Corollary 2 settles the previously missing geometric sharpness edge for these H-type subcases. It does not decide the two strong-type endpoint questions already left open by the source’s positive theorem.

5 Verification, limitations, and novelty status

The reusable script `code/verify_exponents_and_clifford.py` symbolically checks the Q_3, Q_4 substitutions and exactly checks concrete quaternionic and octonionic signed-permutation matrices satisfying (12). These finite checks support, but do not replace, the proof. The accompanying `verification.md` records the command and scope audit.

The construction requires an invariant $(m + 1)$ -block and a tilt annihilating its orthogonal complement. For a general Métivier pencil, $\mathrm{span}\{u, J_1 u, \dots, J_m u\}$ need not be invariant because the vectors $J_i J_j u$ introduce new directions. A moving-block attempt meets the same obstruction as a failure of involutivity of the distribution $\mathrm{span}\{J_1 u, \dots, J_m u\}$ on the sphere. No claim beyond the stated hypotheses is made.

A bounded search of the run indexes and source corpus used the exact source id and phrases involving $Q_3 Q_4$, H-type groups, Métivier sharpness, and center dimension $m \geq 2$. The later general two-step result arXiv:2309.07725 [4] was also checked; it treats the diagonal global maximal problem and still describes the off-diagonal improving problem for $m \geq 2$ as complicated. Two external exact-phrase searches found no statement of the theorem above. This is a bounded check only, so novelty confidence remains provisional.

Human-review recommendation. Check first the cap exponent (9), the vanishing of the mixed bilinear terms, and the cited real Clifford-module dimensions. If those pass, the result appears ready as a substantial partial answer to Remark 1.2(iv).

References

- [1] J. Roos, A. Seeger, and R. Srivastava, *Lebesgue space estimates for spherical maximal functions on Heisenberg groups*, Int. Math. Res. Not. IMRN 2022, no. 24, 19222–19257; arXiv:2103.09734v2.
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- [3] H. B. Lawson, Jr. and M.-L. Michelsohn, *Spin Geometry*, Princeton Mathematical Series 38, Princeton University Press, 1989, Chapter I.
- [4] J. Ryu and A. Seeger, *Spherical maximal functions on two step nilpotent Lie groups*, arXiv:2309.07725.